Descriptive stats:

It consists of organizing and summarising the data**.**

They provide simple summaries of the data. Using graphical analysis and Numerical Analysis.

With descriptive statistics, you are simply describing what is or what the data shows

There are two ways to describe the data as below

1) Numerical OR Stats Method
2) Graphical

Based variable or feature it is also divided in to two parts

1) **Univariate**
2) **Bivariate**

Type of Analysis

- Numerical OR Stats Method
- Graphical Method

Based variable or feature it is also divided in to two parts

- Univariate
- Bivariate

Numerical OR Stats Method

- Univariate Analysis
- Bivariate analysis

- Univariate Analysis
    1) Measures of Central Tendency
    2) Measures of Dispersion
    3) Measure of Location
    4) Measures of Distribution Shape
- Bivariate analysis (Tow variable)
    1) Measures of Association
        - Pearson correlation,
        - Spearman rank correlation

1) Data Transformation(Scalling)
    1) Standardization
    2) Normalization

Graphical Methods

- **Univariate Analysis**
- **Bivariate analysis**

1) Univariate Representation

1) Histogram
2) Bar chart
3) Pie chart
4) Box plot

2) Bivariate Representation

1) Scatter plot
2) Line graph (time series vs variable)
3) Bubble chart

Measure of central tendency

a. Mean

b. Median

c. Mode

Each of these measures describes a different indication of the typical or **central value** in the distribution.

Mean: Find out the **Average** of the data

Find mean of [2,3,4,5,6]

$$x = \frac{2+3+4+5+6}{5}$$

$$x = \frac{20}{5}$$

$$x = 4$$

Median: It will find the **central Element.**

Median for Even:

If the Number of elements are even the find the average of the central elements.

Ex [3,4,**5,6**,8,100]

Median for Even

=5+6/2

=11/2

=5.6

Median for Odd:

If the Number of elements are odd then find the average of the central element.

Ex [3,4,**5,**6,120]

We can directly take the central element as the final value for the Median which is 5 in this case.

Mode: Mode is the **most frequently** occurring element. Usually applied on categorical Data

Example1: [2,2,2,3,4,4,5,5,5]

2 and 5 is a mode for the above case

Example2: [2,2,2,3,4,4,5,5,7,8,8]

2 is the mode in this case since 2 is occurring most of the time.

Measure of dispersion(Variability):

The measures of central tendency are not adequate to describe data. Two data sets can have the same mean but they can be entirely different. Thus to describe data, one needs to know the extent of variability. This is given by the measures of dispersion.

1) **Variance**
2) **Standard deviation**
3) **Interquartile range (IQR)**

Variance:

Variance measures the average squared deviation of each data point from the **mean**. Variance talks about the spread of data.

Variance is a measure of dispersion that indicates how far a set of numbers is spread out from their average (mean) value. A low variance indicates that data points cluster closely around the mean, while a high variance indicates that data points are widely distributed.

$$\text{x } [1,2,3,4,5]$$

$$x = \frac{\sum_{I=1}^{N}(X_I - u)^2}{N-1}$$

$$x = \frac{(1-3)^2+(2-3)^2+(3-3)^2+(4-3)^2+(5-3)^2}{5-1}$$

$$x = \frac{(1-3)^2+(2-3)^2+(3-3)^2+(4-3)^2+(5-3)^2}{4}$$

$$x = \frac{(-2)^2+(-1)^2+(0)^2+(1)^2+(2)^2}{4}$$

$$x = \frac{4+1+0+1+4}{4}$$

$$x = \frac{10}{4}$$

$$x = 2.5$$

There are two types of variance,

Population Variance

Sample Variance.

Sample Variance(σ^2):

When the number of observations increases then a few data points are selected that can represent the entire population. These specific data points form a sample and the variance calculated on this data is called the sample variance.
Calculate the Population variance for the below dataset.
Consider the above(X) sample data is taken from the population data

X [1,2,3,4,5]

$$x = \frac{\sum_{I=1}^{N}(X_I - u)^2}{N}$$

$$x = \frac{(1-3)^2+(2-3)^2+(3-3)^2+(4-3)^2+(5-3)^2}{5}$$

$$x = \frac{(-2)^2+(-1)^2+(0)^2+(1)^2+(2)^2}{5}$$

$$x = \frac{4+1+0+1+4}{5}$$

$$x = \frac{10}{5}$$

$$x = 2$$

Population Variance(σ^2):

When the data size is small then the population variance is calculated on the actual population dataset.

Calculate the Population variance for the below dataset.

X [1,2,3,4,5]

$$x = \frac{\sum_{I=1}^{N}(X_I - u)^2}{N}$$

$$x = \frac{(1-3)^2+(2-3)^2+(3-3)^2+(4-3)^2+(5-3)^2}{5}$$

$$x = \frac{(-2)^2+(-1)^2+(0)^2+(1)^2+(2)^2}{5}$$

$$x = \frac{4+1+0+1+4}{5}$$

$$x = \frac{10}{5}$$

$$x = 2$$

Dataset: 2, 4, 6, 8

- Step 1: Calculate the sample mean ($\bar{x}$)

$$\bar{x} = \frac{2+4+6+8}{4} = \frac{20}{4} = 5$$

- Step 2: Find the squared deviation for each data point $(x_i - \bar{x})^2$
  - $(2-5)^2 = (-3)^2 = 9$
  - $(4-5)^2 = (-1)^2 = 1$
  - $(6-5)^2 = (1)^2 = 1$
  - $(8-5)^2 = (3)^2 = 9$
- Step 3: Sum the squared deviations

$$\sum (x_i - \bar{x})^2 = 9 + 1 + 1 + 9 = 20$$

- Step 4: Divide by $n-1$ (Bessel's Correction)

$$s^2 = \frac{20}{4-1} = \frac{20}{3} \approx 6.67$$

Standard Deviation(σ):

It is statistical measure of the amount of **dispersion** in a set of values. It measures how spread out the values in a dataset are from the mean (average) value. A low standard deviation indicates that the values tend to be close to the mean, while a high standard deviation indicates that the values are spread out over a wider range.

STD

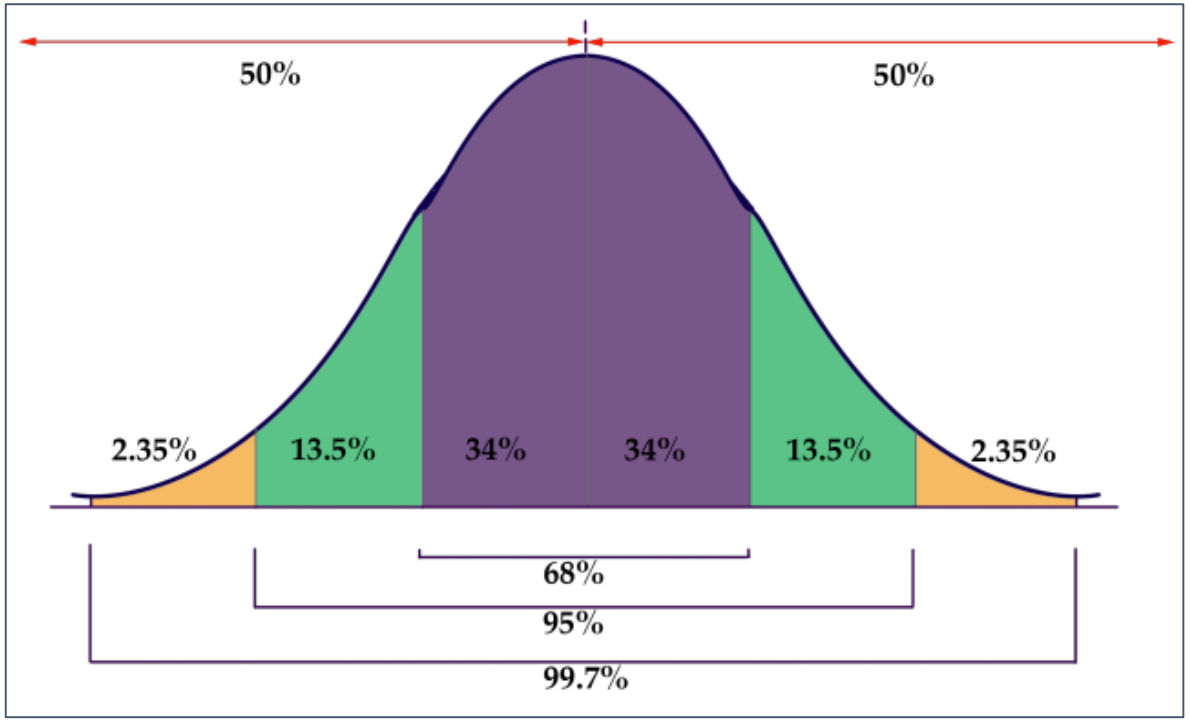
$$\boldsymbol{x = [1, 2, 3, 4, 5]}$$

$$x = \sqrt{\frac{\sum_{I=1}^{N}(X_I - u)^2}{N}}$$

$$x = \sqrt{\frac{(1-3)^2+(2-3)^2+(3-3)^2+(4-3)^2+(5-3)^2}{5}}$$

$$x = \sqrt{2}$$

$$x = 1.41$$



Normally Disturbed Data.

Sample STD

$x = [1,2,3,4,5]$

$$x = \sqrt{\frac{\sum_{I=1}^{N}(X_I - u)^2}{N-1}}$$

$$x = \sqrt{\frac{(1-3)^2+(2-3)^2+(3-3)^2+(4-3)^2+(5-3)^2}{4}}$$

$x = \sqrt{2.5}$

$x = 1.581$

Population STD

$x = [1,2,3,4,5]$

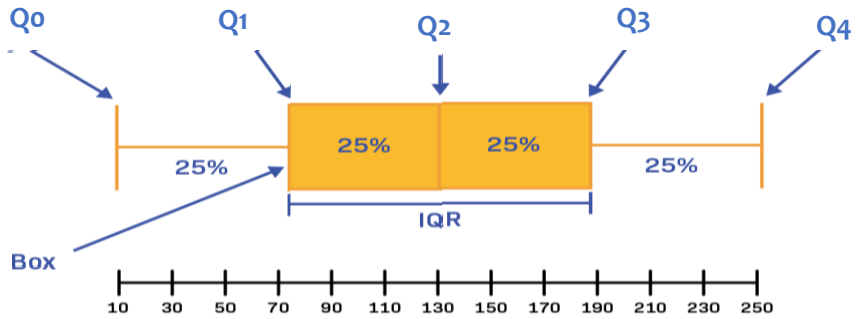
$$x = \sqrt{\frac{\sum_{I=1}^{N}(X_I - u)^2}{N}}$$

$$x = \sqrt{\frac{(1-3)^2+(2-3)^2+(3-3)^2+(4-3)^2+(5-3)^2}{5}}$$

$x = \sqrt{2}$

$x = 1.41$

Inter Quartile Range(IQR)



Q1 = 0.25
Q3 = 0.75

Inter Quartile Range(IQR) = Q3 - Q1
Lower Fence = Q1 - 1.5 * IQR
Higher Fence = Q3 + 1.5 * IQR

Inter Quartile Range(IQR) = 190 - 72
Inter Quartile Range(IQR) = 118

Lower Fence = 72 - 1.5 * 118
Lower Fence = -105

Higher Fence = Q3 + 1.5 * IQR
Higher Fence = 367

Measures of Position (Location):

Measures of position (also called measures of location) are statistical tools that tell us **where a particular value stands within a dataset**. They help us understand the **relative standing** of a data point compared to the rest of the data.

1) **Percentile**
2) **Quartile**
3) **Z-scores**

Percentile:

It tells you what percentage of data points are below or equal to a given value.

1) Sort the value
2) Use below equation to calculate the Percentile

$$x = \frac{n}{m} * 100$$

$n = Number\ of\ Value\ till\ x$
$m = Total\ number\ of\ Value$

Calculate the percentile of value 8 on below data set

x = [5, 4, 6, 10, 1, 7, 8, 3, 9, 2]

x = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]

$$x = \frac{n}{m} * 100$$

$$x = \frac{7}{10} * 100$$

$$x = 70\ \%$$

Calculate the percentile of below data set

x = [77,83,97,85,74,70,98,91,94,78,65,40,69,72,94,60,76,49]

x = [40,49,60,65,69,70,72,74,76,77,78,83,85,91,94,94,97,98]

$$x = \frac{n}{m} * 100$$

Percentile of number 49

$$x = \frac{1}{18} * 100$$

$$x = 5.56\%$$

Percentile of number 77

$$x = \frac{9}{18} * 100$$

$$x = 50.00\ \%$$

Quartile

Quartiles are values that divide a dataset into **four equal parts.** They help us understand how the data is distributed and how much the data varies.

There are **three quartiles:**

1. First Quartile (Q1)

Q1 is the **25th percentile.**

It is also called the **Lower Quartile.**

It means **25% of the data values are below Q1.**

2. Second Quartile (Q2)

Q2 is the **50th percentile.**

It is also called the **Median.**

It divides the data into **two equal parts.**

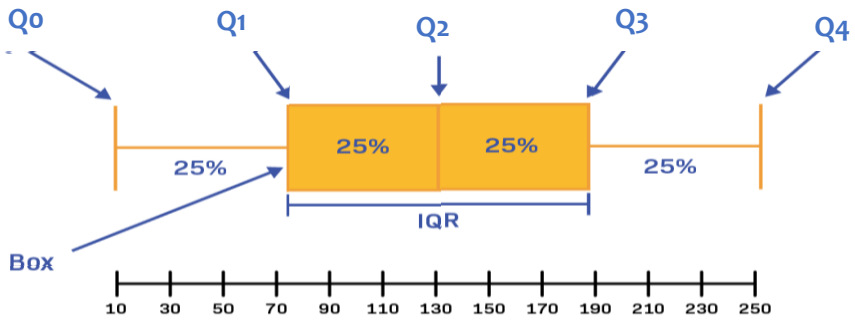
50% of the data is below Q2 and 50% is above Q2.

3. Third Quartile (Q3)

Q3 is the **75th percentile.**

It is also called the **Upper Quartile.**

It means **75% of the data values are below Q3,** while the remaining 25% are above it.



$$z - Score = \frac{x - u}{\sigma}$$

$$z - Score = \frac{(1-3)+(2-3)+(3-3)+(4-3)+(5-3)}{1.41}$$

$$z - Score = \frac{(-2)+(-1)+(0)+(1)+(2)}{1.41}$$

$$z - Score = \frac{0}{1.41}$$

$$z - Score = 0$$

STD

$$\boldsymbol{x} = \ [\mathbf{1}, \mathbf{2}, \mathbf{3}, \mathbf{4}, \mathbf{5}]$$

$$x = \sqrt{\frac{\sum_{I=1}^{N}(X_I - u)^2}{N}}$$

$$x = \sqrt{\frac{(1-3)^2+(2-3)^2+(3-3)^2+(4-3)^2+(5-3)^2}{5}}$$

$$x = \sqrt{2}$$

$$x = 1.41$$

Student	Marks	Z-score
A	40	-1.41
B	50	-0.71
C	60	0
D	70	0.71
E	80	1.41

$$\text{Mean} = \frac{40 + 50 + 60 + 70 + 80}{5}$$

$$= \frac{300}{5} = 60$$

$$\mu = 60$$

$$std = \sqrt{\frac{(40-60)^2+(50-60)^2+(60-60)^2+(70-60)^2+(80-6)^2}{5}}$$

$$std = \sqrt{\frac{(-20)^2+(-10)^2+(0)^2+(10)^2+(20)^2}{5}}$$

$$std = \sqrt{\frac{400+100+0+100+400}{5}}$$

$$std = \sqrt{\frac{1000}{5}}$$

$$std = \sqrt{200}$$

$$Std = 14.142$$

$$Std = 14.142$$

Step 3: Calculate Z-score

Formula:

$$z = \frac{X - \mu}{\sigma}$$

For **40**:

$$z = \frac{40 - 60}{14.14} = -1.41$$

For **50**:

$$z = \frac{50 - 60}{14.14} = -0.71$$

For **60**:

$$z = \frac{60 - 60}{14.14} = 0$$

For **70**:

$$z = \frac{70 - 60}{14.14} = 0.71$$

For **80**:

$$z = \frac{80 - 60}{14.14} = 1.41$$

Measures of Distribution Shape:

When analyzing data, measures of distribution shape help us understand the way the data is spread out or clustered. The most common measures of distribution shape include:

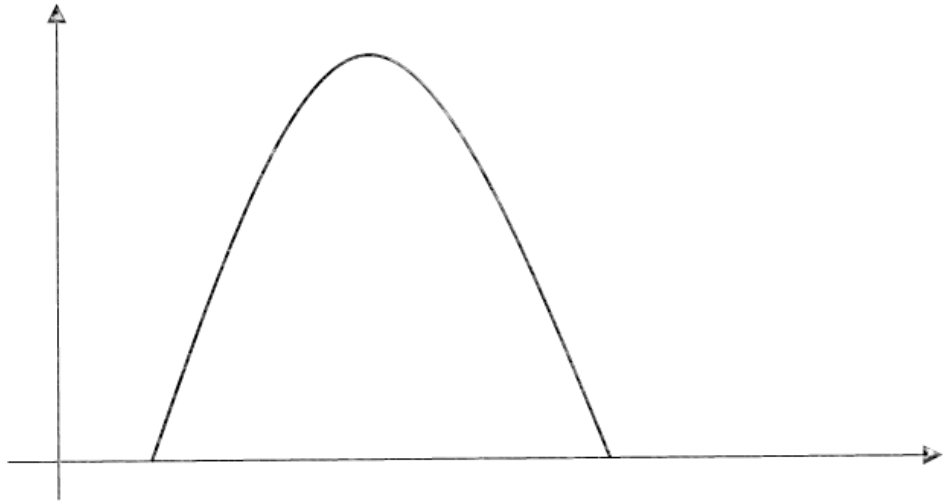
Skewness	Kurtosis
• Left • Right	• Positive • Negative

Skewness:
A measure of the asymmetry of the distribution.

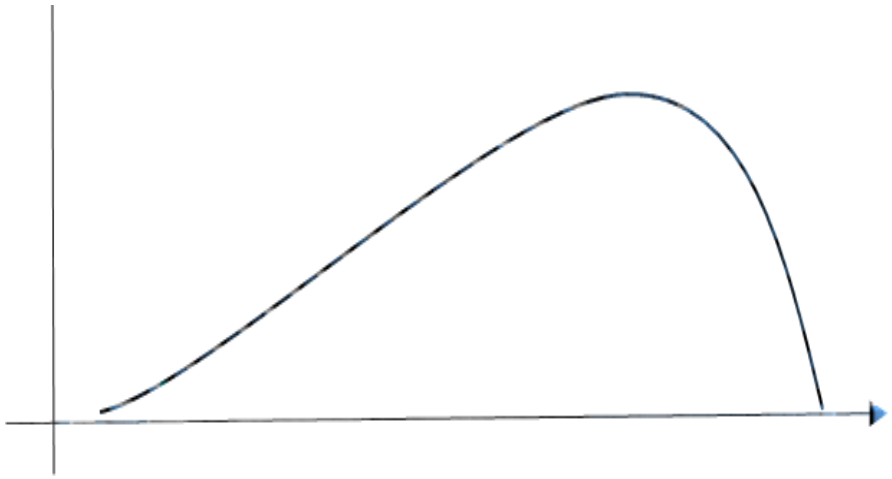
Negative skewness indicates a longer tail on the left side.

Positive skewness indicates a longer tail on the right side of the distribution,

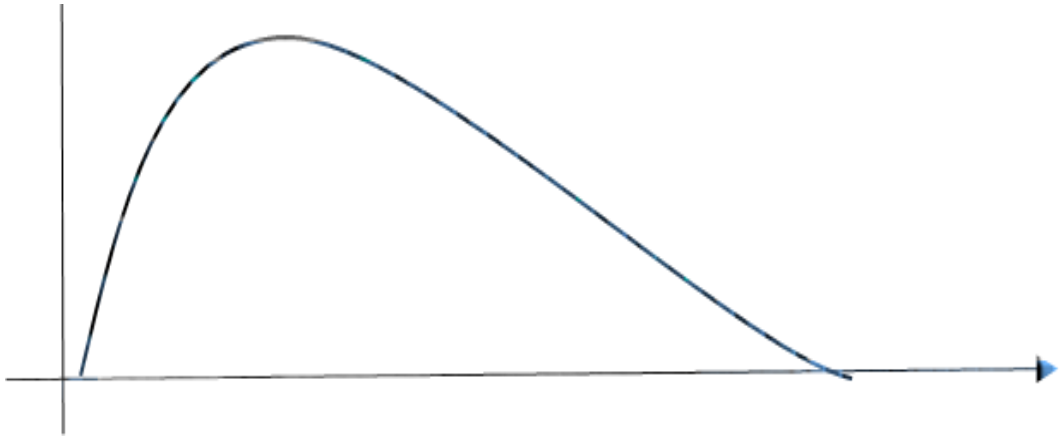
Normal Distribution, no tail on either side



Normally distributed (Gaussian Curve)



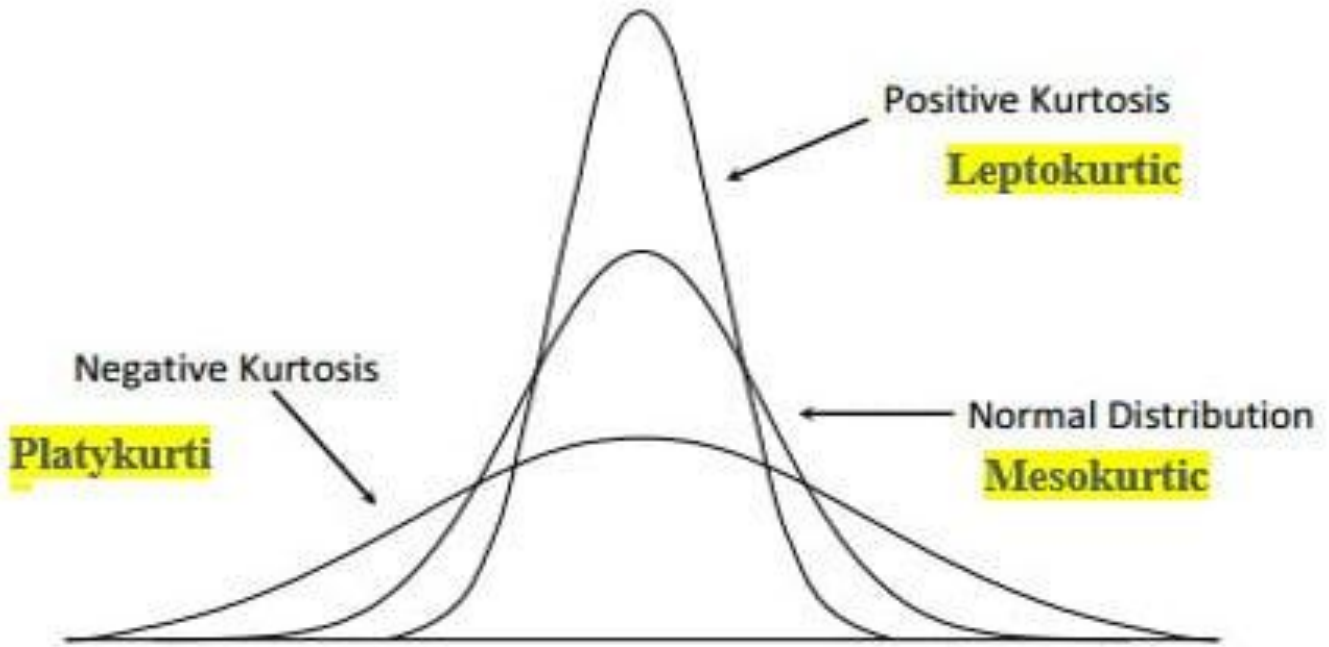
Left(Negative) Skewness



Right(Positive) Skewness

Kurtosis:

- Higher kurtosis indicates sharper, more peaked distributions,
- while lower kurtosis indicates flatter distributions.



Bivariate analysis

Measures of Association

Measures of Association

Measures of association tell you how two variables change together.

Variance:

Measures how **one** variable changes relative to its own mean.

Covariance:

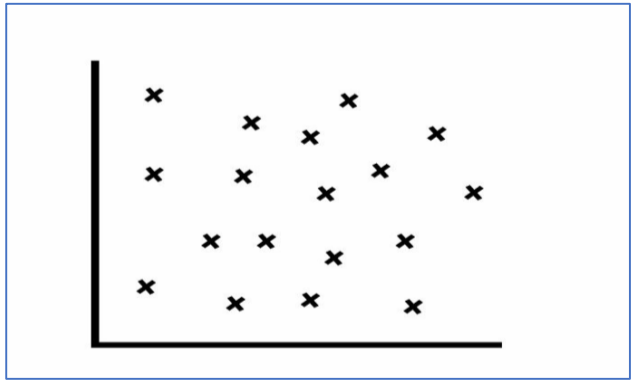
Measures how **two** variables change together relative to their means. (It is literally the "co-" variance).

Correlation:

Normalizes covariance so you can actually judge the strength of that movement on a fixed scale of -1 to +1.

Covariance:

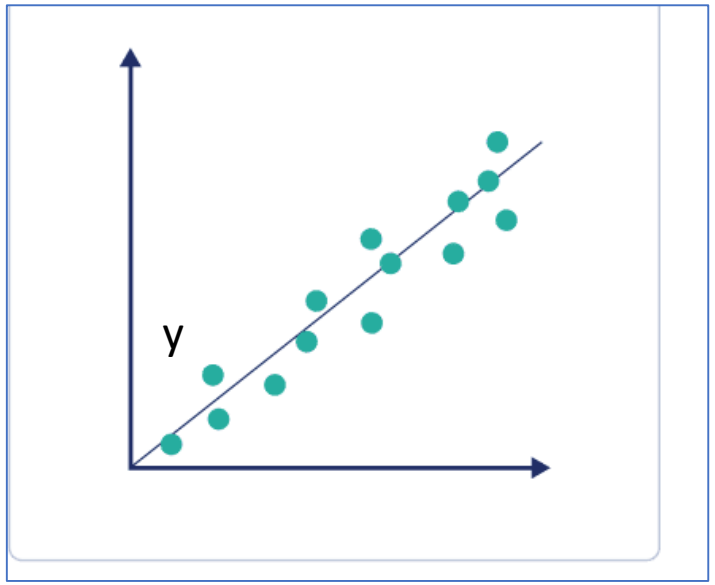
Measures how **two** variables change together relative to their means. (It is literally the "co-" variance). It gives you the Direction



+ Ve **Covariance**

X is Increasing and Y is Increasing

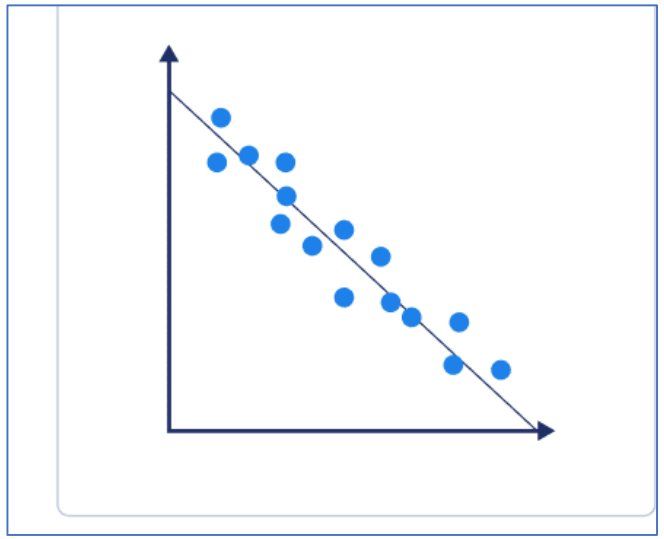
X is Decreasing and Y is Decreasing



-Ve **Covariance**

X is Increasing and Y is Decreasing

X is Decreasing and Y is Increasing



Covariance o:

No relation between X and Y

Covariance:

+ Ve Covariance:

Student	X(Hrs)	Y(Marks
A	1	2
B	2	4
C	3	6
D	4	8
E	5	10

X – 3
-2
-1
0
1
2

Y – 6
-4
-2
0
2
4

(X–3)*(Y–6)
8
2
0
2
8
20

$$cov = \frac{\sum(x - x\,min) * (y - y\,min)}{n - 1}$$

$$cov = \frac{20}{4}$$

$$cov = 5$$

- Ve Covariance:

Student	Tv Hrs (x)	Marks (x)	(X – 3)	Y – 3	(X-3)*(Y-3)
A	1	5	-2	-4	-4
B	2	4	-1	-2	-1
C	3	3	0	0	0
D	4	2	1	2	-1
E	5	1	2	4	-4
					-10

$$cov = \frac{\sum(x - x\,min) * (y - y\,min)}{n - 1}$$

$$cov = \frac{-10}{4}$$

$$cov = -2.50$$

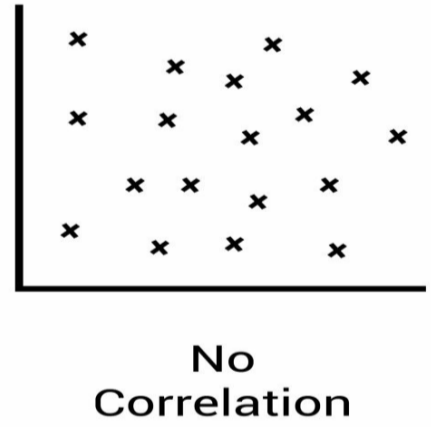
Correlation :

It is used to quantify the relationship or association between variables in a dataset.

A measure of how much two variables change together. It indicates the Direction and Strength to which two variables change(vary) together.

Pearson correlation coefficient:

Spearman correlation:

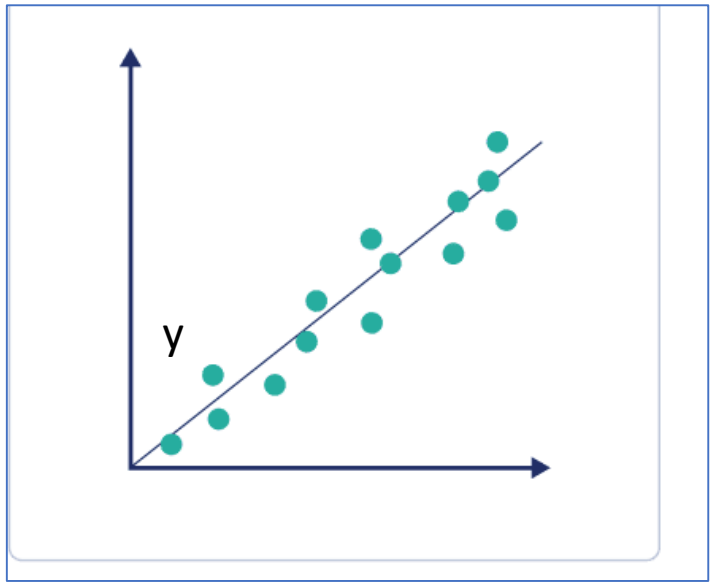


No Correlation

+ Ve **Correlation**

X is Increasing and Y is Increasing

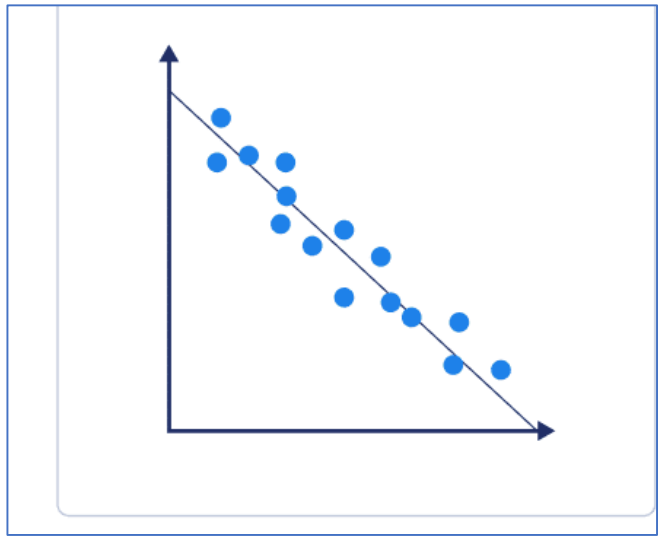
X is Decreasing and Y is Decreasing



-Ve **Correlation**

X is Increasing and Y is Decreasing

X is Decreasing and Y is Increasing



Correlation 0:

No relation between X and Y

Pearson correlation coefficient:

Restricting the values between(-1 to +1)
The more the values toward the +1 more positively correlated it is and vice-versa.

+ Ve correlation
-Ve Correlation

$$A = \frac{cov(X, Y)}{SD_X * SD_Y}$$

+ Ve Correlation :

Student	X(Hrs)	Y(Marks
A	1	2
B	2	4
C	3	6
D	4	8
E	5	10

$$A = \frac{cov(X,Y)}{SD_X * SD_Y}$$

$$A = \frac{5}{1.5811 * 3.16}$$

$$A = +1$$

- Ve Correlation :

Student	Study Hrs (X)	TV Hours (Y)
A	1	5
B	2	4
C	3	3
D	4	2
E	5	1

$$A = \frac{cov(X,Y)}{SD_X * SD_Y}$$

$$A = \frac{-2.5}{1.581 * 1.581}$$

$$A = -1$$

Spearman correlation:

It is used only for non-linear data.

It assigns ranks

Spearman's rank correlation measures the strength and direction of association between two ranked variables.

$$r = 1 - \frac{6 \sum d^2}{n(n^2 - 1)}$$

+ ve Spearman Rank correlation

Students	Marks of Statistics (X)	Rank of X	Marks of English (Y)	Rank of Y	d	d2
F	25	8	40	6	2	4
C	35	7	51	5	2	4
B	56	5	61	4	1	1
E	72	3	72	3	0	0
A	83	2	90	1	1	1
D	99	1	98	1	0	0

$$r = 1 - \frac{6\sum d^2}{n(n^2-1)}$$

$$r = 1 - \frac{6 * 10}{6(6^2-1)}$$

$$r = 1 - \frac{60}{6(36-1)}$$

$$r = 1 - \frac{60}{6(35)}$$

$$r = 1 - 0.2857$$

$$r = 0.7143$$

- Ve Spearman Rank correlation

Car Model	Age (Years)	Rank of Age (Rx)	Resale Price (₹)	Rank of Price (Ry)	(Rx−Ry)= d	d2
Model A	1	1	9,00,000	5	1 - 5 = -4	16
Model B	2	2	10,50,000	6	2 - 6 = -4	16
Model C	3	3	6,00,000	3	3 - 3 = 0	0
Model D	4	4	7,50,000	4	4 - 4 = 0	0
Model E	5	5	3,00,000	1	5 - 1 = 4	16
Model F	6	6	4,50,000	2	6 - 2 = 4	16

$$r_s = 1 - \frac{6 \sum d^2}{n(n^2 - 1)}$$

$$r_s = 1 - \frac{6(64)}{6(36 - 1)}$$

$$r_s = 1 - \frac{384}{210}$$

$$r_s = 1 - 1.8286 = -0.8286$$

Correlation coefficient (R)	Interpretation
1 > R > 0.75	High positive correlation
0.75 > R > 0.5	Moderate positive correlation
0.5 > R > 0.1	Weak positive correlation
0.1 > R > -0.1 (or approximately **R ≈ 0**)	No correlation

Correlation coefficient (R)	Interpretation
-0.1 > R > -0.5	Weak negative correlation
-0.5 > R > -0.75	Moderate negative correlation
-0.75 > R > -1	High negative correlation

Graphical Analysis

Data visualization methods split based on whether you are examining a single variable (univariate) or exploring relationships between two variables (bivariate).

Univariate Graphs (Single Variable) Focuses on the distribution, central tendency, and spread of one feature at a time.

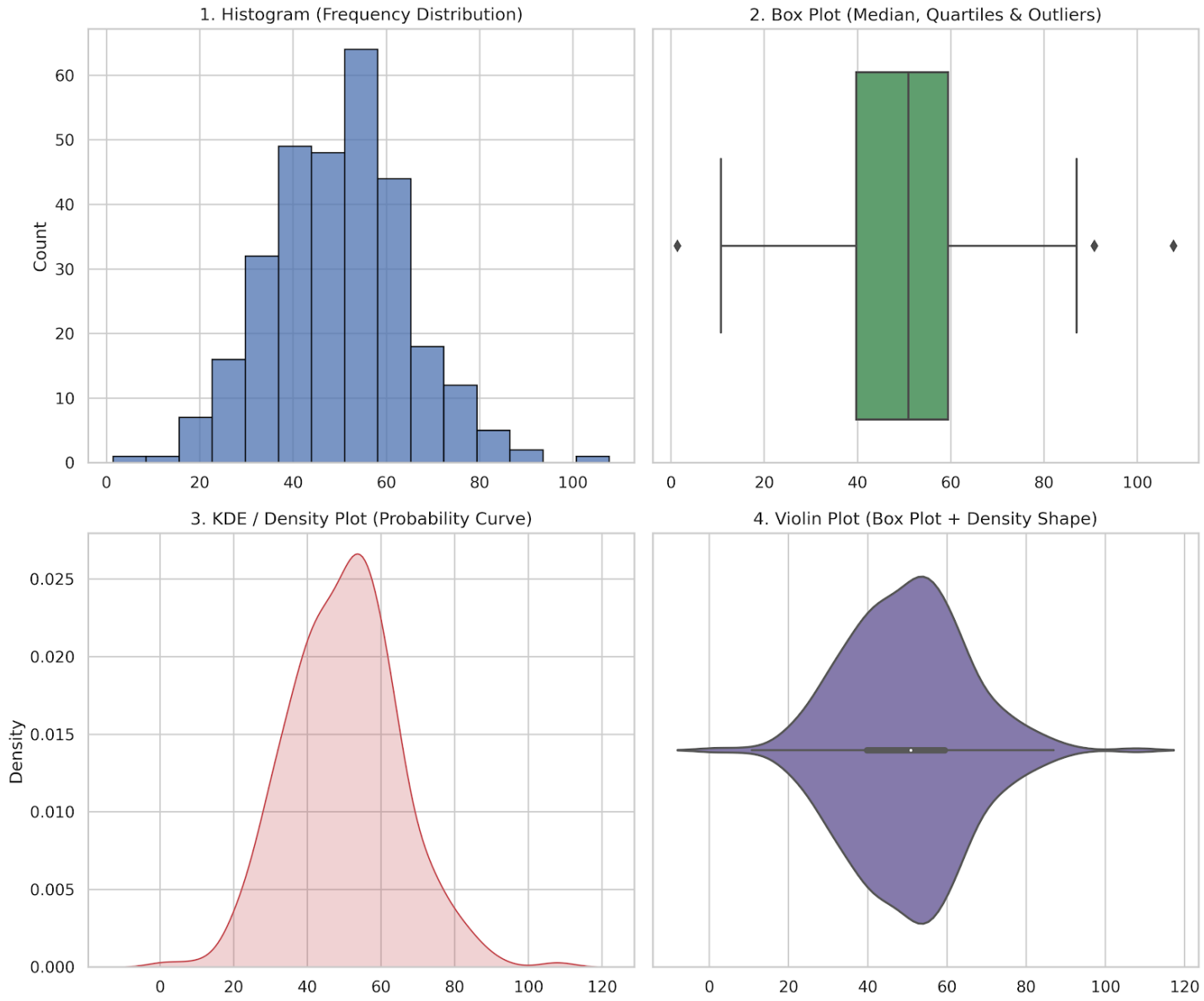
- **Numerical Data:**
- **Categorical Data:**

Bivariate Graphs (Two Variables) Examines correlations, distributions between a pair of features.

- **Numerical vs. Numerical:**
- **Categorical vs. Categorical:**
- **Numerical vs. Categorical:**

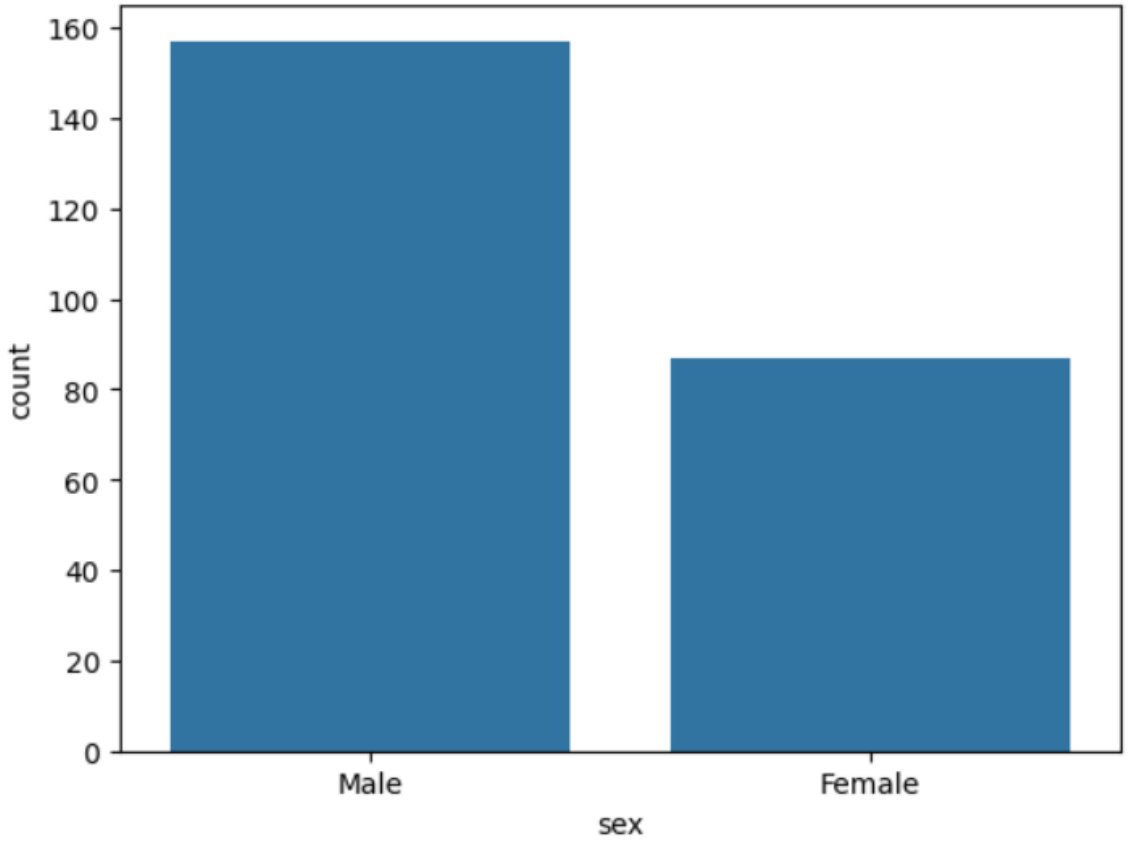
Numerical Data:

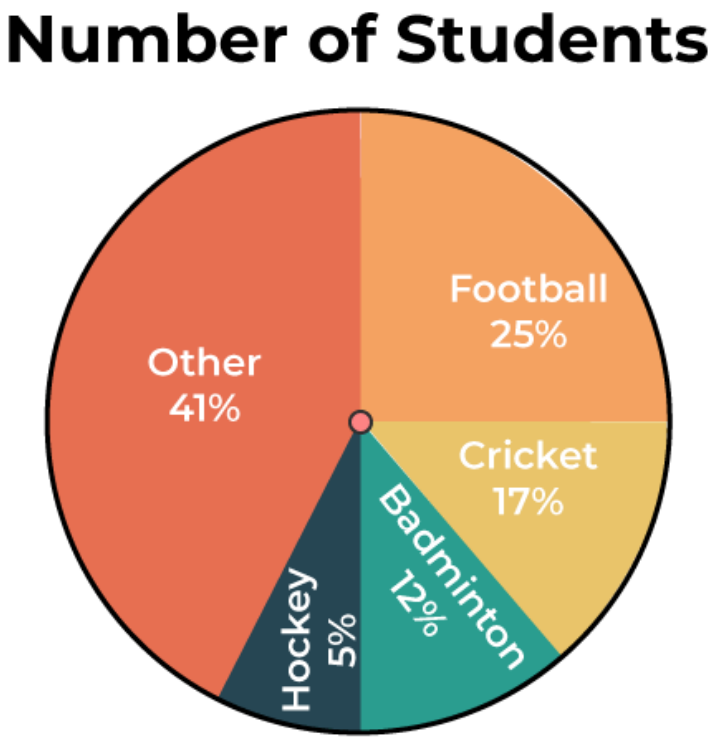
- **Histogram:** Splits continuous numerical data into equal-width bins to show frequency distribution.
- **Box Plot:** Highlights the median, quartiles, and outliers using interquartile ranges (IQR).
- **KDE / Density Plot:** Smoothes data values into a continuous probability density curve.
- **Violin Plot:** Combines a box plot with a kernel density estimation to show distribution shape and multimodality.



Categorical Data:

- **Bar Chart / Count Plot:** Displays the absolute or relative frequency counts of discrete categories.
- **Pie Chart:** Represents proportions of a whole across a limited number of distinct categories.



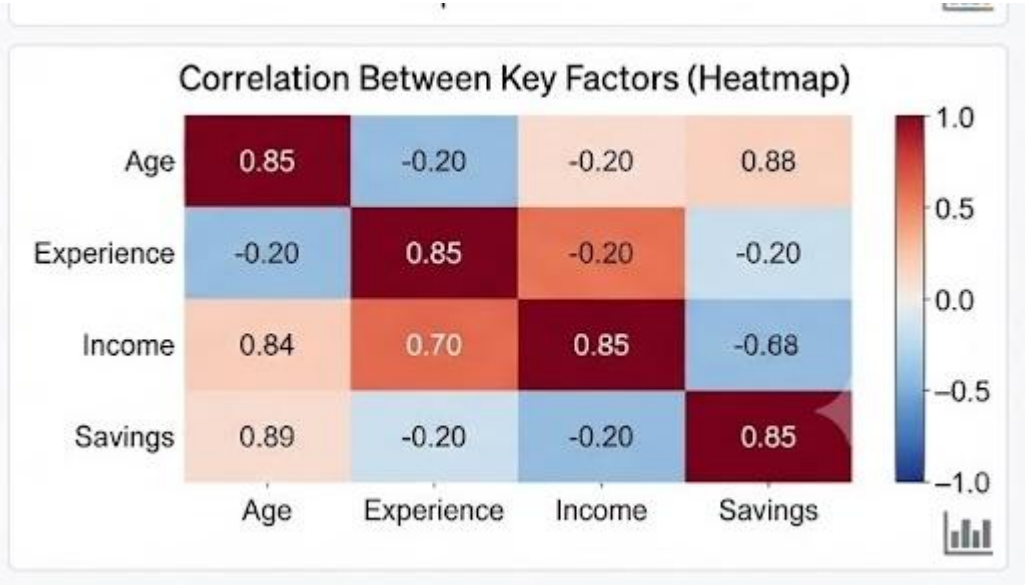


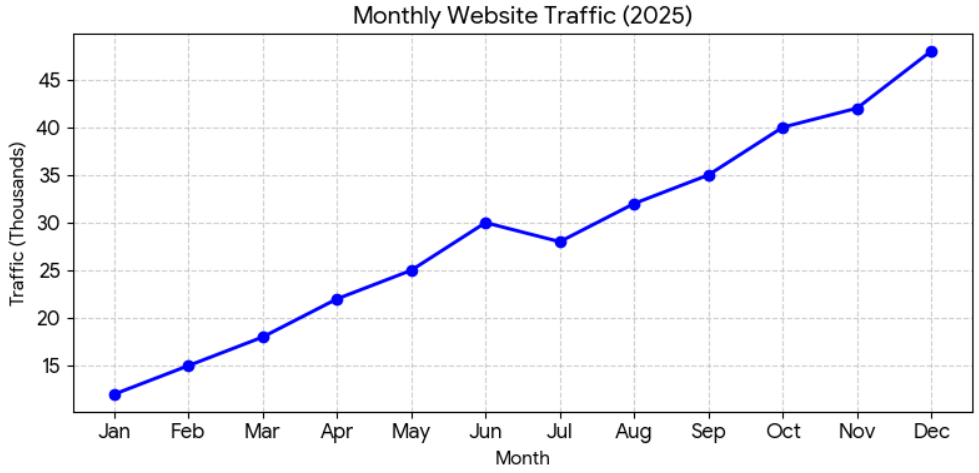
Bivariate Graphs (Two Variables) Examines correlations, dependencies, and comparative distributions between a pair of features.

Numerical vs. Numerical:

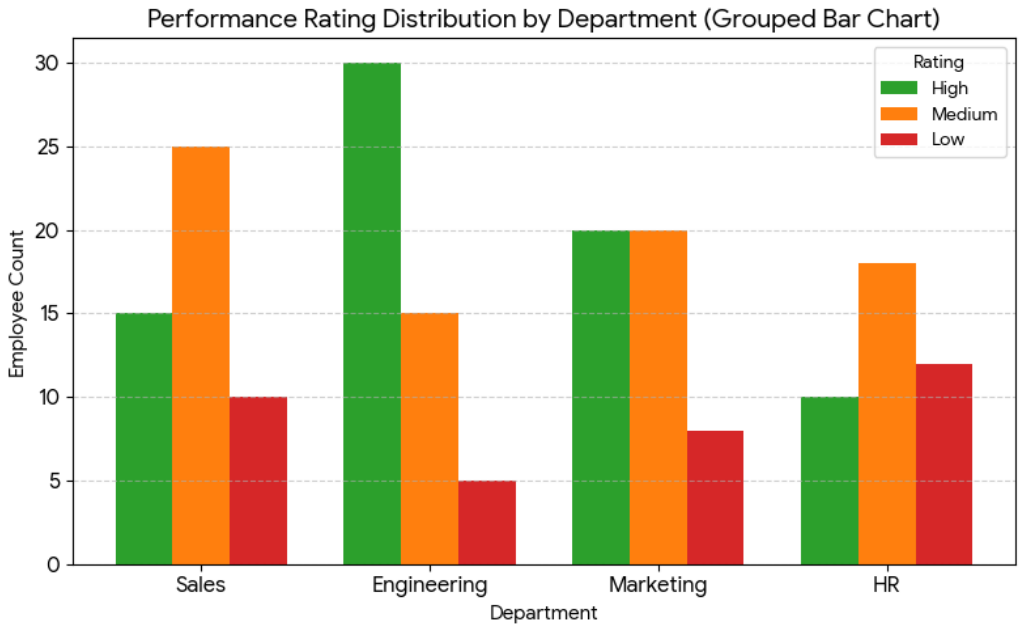
- **Scatter Plot:** Plots individual data points on X and Y axes to reveal linear or non-linear correlations and clusters.
- **Line Chart:** Connects sequential data points, typically tracking numerical changes over time or continuous intervals.
- **Correlation Heatmap:** Displays pairwise correlation coefficients in a matrix format.

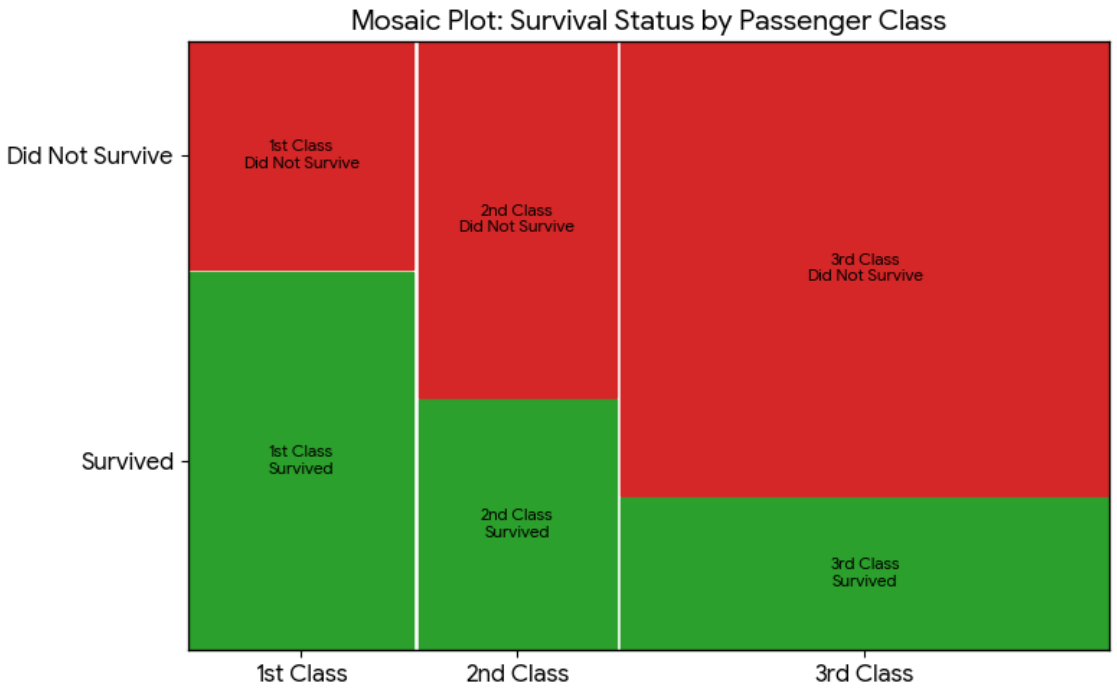






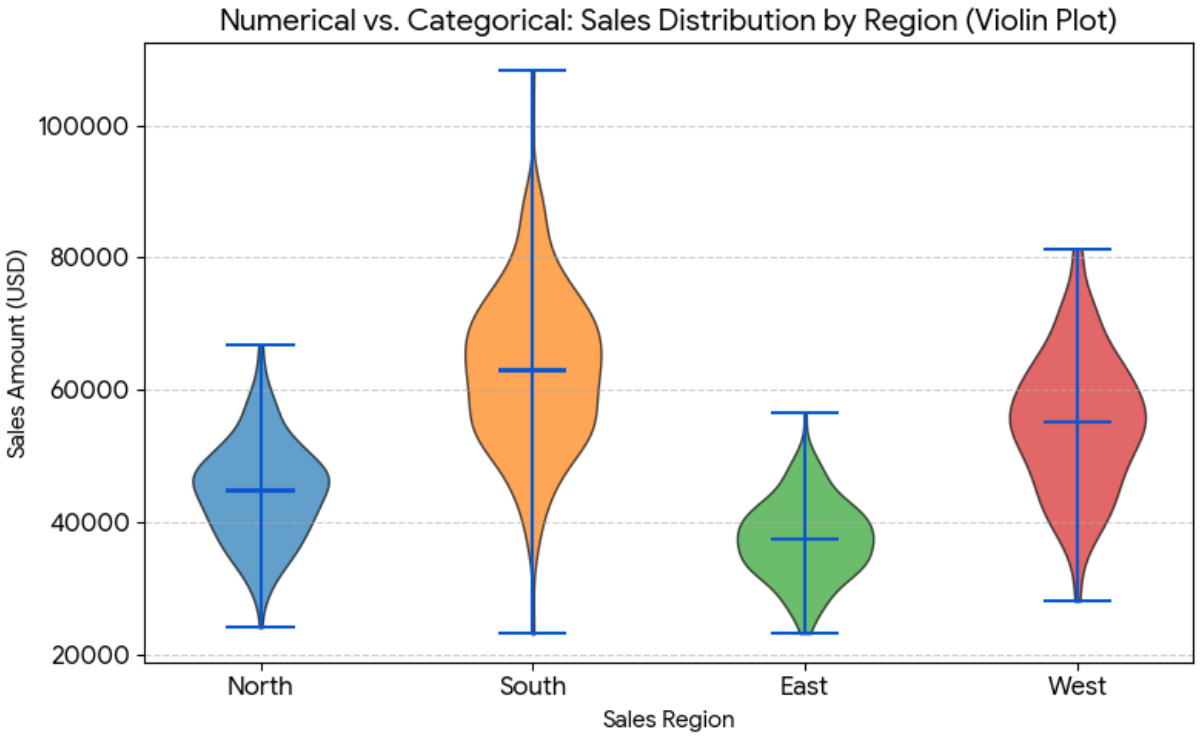
- **Categorical vs. Categorical:**
  - **Grouped / Stacked Bar Chart:** Compares sub-categories within primary groups using side-by-side or stacked bars.
  - **Mosaic Plot:** Visualizes cross-tabulated categorical data using proportional rectangular areas.

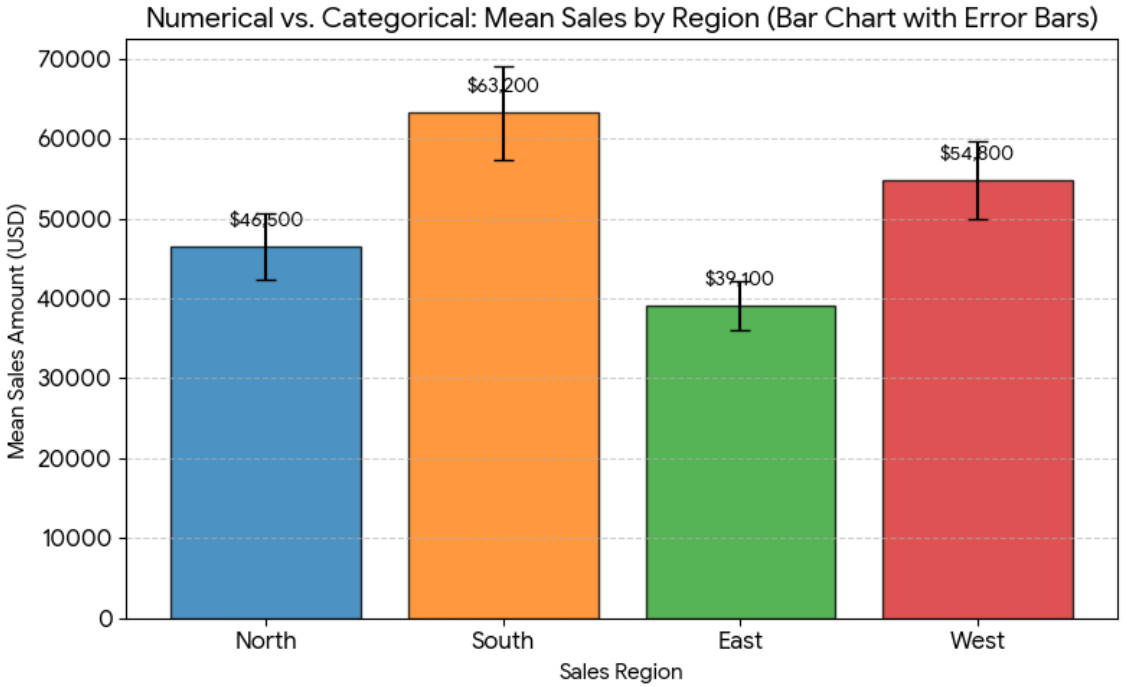




Numerical vs. Categorical:

- **Grouped Box / Violin Plot:** Compares the distribution, median, and spread of a numerical variable across multiple discrete categories.
- **Bar Chart of Means:** Shows aggregated values (such as means, medians, or sums) for a numerical metric sliced by categorical groups.
- **Swarm / Strip Plot:** Overlays individual jittered data points on top of categorical distribution plots to display underlying sample sizes.





THE END